

Allan Vikiru

Data Science | Applied AI | Internet of Things

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Skills and Competencies

- Software Engineering
- Machine Learning
- Hardware Prototyping
- Spatial Data Analysis
- IoT Prototyping
- Data Engineering
- Web and Cloud Development
- Object-Oriented Programming
- Commercial and Academic Presentations
- Technical Writing
- Project Management
- Stakeholder Engagement

Languages: C++, JavaScript, LATEX, MATLAB, Python, R

Software: Arduino, InfluxDB, Git, Django, Heroku, Microsoft Power BI, QGIS, Streamlit

Education

MSc. Information Technology (Business Intelligence & Data Analytics Option) [2024]

Strathmore University - Nairobi, Kenya

Thesis Title: A Predictive Model for Determining Vegetable-Derived Macronutrients for a Diabetic Patient

BSc. Informatics & Computer Science [2021]

Strathmore University - Nairobi, Kenya

Final Year Project Title: Determining the Risk of Fireground Injuries among Firefighters using IoT

Work Experience

Tutorial Fellow – School of Computing and Engineering Sciences, Strathmore University [July 2024 – Present]

- Teaching at least 250 students on Artificial Intelligence, Embedded Systems & IoT, Information Security, Data Governance and Cloud Computing.
- Supervising up to 25 students on projects related to challenge-driven education, deep learning and Internet of Things.

Research Coordinator – SCES Makerspace Lab, Strathmore University [July 2023 – Present]

- Continually engaging a team of 6 researchers and at least 10 undergraduate and graduate students in developing computing and engineering projects.
- Co-lead of the **Conservation Technology Research Hub**. Key tasks include internal and external stakeholder engagement, data collection and analysis, device prototyping, grant proposal writing and conducting team meetings.

Graduate Assistant – School of Computing & Engineering Sciences, Strathmore University [March 2022 – June 2024]

- Developed a research proposal on using machine learning and IoT to detect logging activities that received an award of USD 139,500.
- Contributor to three research papers on deep learning and cloud computing.

Software Developer Intern – IoT & Wireless Networks Lab, @iLabAfrica [January 2022 - March 2022]

- Designed and implemented a monitoring dashboard in Vue.js and Node.js and an authentication system in Keycloak for a LoRaWAN-based research project.
- Designed and developed breadboard model devices to model LoRaWAN connectivity.

Publications

Vikiru A. & Khakata E. (in press). A Wearable Sensor System for Monitoring the Risk of Fireground Injuries among Firefighters.

11th International Congress on Information and Communication Technology.

Ayankoso, S., Wang, Z., Shi, D., Yang, W., **Vikiru, A.**, Kamau, S., Muchiri, H., & Gu, F. (2024). Development of Long-Range, Low-Powered and Smart IoT Device for Detecting Illegal Logging in Forests. *Journal of Dynamics, Monitoring and Diagnostics*.

<https://doi.org/10.37965/jdmd.2024.550>

Simiyu, D., **Vikiru, A.**, Muchiri, H., Gu, F., & Butime, J. (2024). A Chainsaw-Sound Recognition Model for Detecting Illegal Logging Activities in Forests. In A. D. Ball, H. Ouyang, J. K. Sinha, & Z. Wang (Eds.), *Proceedings of the UNified Conference of DAMAS, InCoME and TEPEM Conferences (UNified 2023)* (pp. 797–806). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-49421-5_65

Vikiru, A. (2023, March 18). *Market Basket Analysis using Apriori Algorithm*. <https://allanvikiru.github.io/MarketBasketAnalysis/>

Vikiru, A., Muiruri, M., & Ateya, I. (2023). *An Overview on Cloud Distributed Databases for Business Environments*

(arXiv:2301.10673). arXiv. <https://doi.org/10.48550/arXiv.2301.10673>

Conferences & Workshops

Delegate – GDH Partner Days on Challenge Driven Education

Royal Institute of Technology (KTH) - Stockholm, Sweden

March 2026

Presenter - A Wearable Sensor System for Monitoring the Risk of Fireground Injuries among Firefighters

11th International Congress on Information and Communication Technology - London, United Kingdom

February 2026

Delegate – Microsoft Faculty Skilling and Immersion Program (Cohort 3 – AI-Oriented Education)

Microsoft ADC & Microsoft Leap - Nairobi, Kenya

June 2025

Delegate - 5th International Conference on Smart Sensors and Application

Universiti Teknologi Malaysia (UTM) - Penang, Malaysia

September 2024

Project Administrator - Curbing Illegal Logging Patterns Using Sound-Based Detection Techniques

Strathmore University (Kenya), Université de Neuchâtel (Switzerland), University of Huddersfield (United Kingdom), Kenya Forestry Research Institute (Kenya)

July 2023 - September 2024

Delegate – CMU Innovation Summit 2024

Carnegie Mellon University Africa - Kigali, Rwanda

January 2024

Presenter – Fundamentals of Raspberry Pi 400

Strathmore University - Nairobi, Kenya

November 2023

Delegate - International Summer School in Cyber Security and Digital Forensics

Brno University of Technology - Brno, Czech Republic

June 2023